

Database System Engineering and Distributed Backend Development Course

Code: 25CS1302E

ROLL NO : 2520030255

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WEEK-2

```
mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
mysql> USE bookflow_db;
Database changed
mysql> CREATE TABLE Books (
  -> book_id INT PRIMARY KEY,
  -> title VARCHAR(100) NOT NULL,
  -> isbn VARCHAR(20) UNIQUE,
  -> published_year INT CHECK (published_year < 2027)
  -> );
ERROR 1050 (42S01): Table 'books' already exists
mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES
  -> (1, 'The Great Gatsby', '9780743273565', 1925),
  -> (2, 'To Kill a Mockingbird', '9780061120084', 1960),
  -> (3, '1984', '9780451524935', 1949);
ERROR 1062 (23000): Duplicate entry '1' for key 'books.PRIMARY'
mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title          | isbn          | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist  | 9780061122415 | 1988           |
| 2       | Clean Code     | 9780132350884 | 2008           |
| 3       | Atomic Habits  | 9780735211292 | 2018           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> CREATE TABLE Members (
  -> member_id INT PRIMARY KEY,
  -> full_name VARCHAR(100),
  -> email VARCHAR(100) UNIQUE
  -> );
ERROR 1050 (42S01): Table 'members' already exists
mysql> INSERT INTO Members (member_id, full_name, email) VALUES
  -> (101, 'John Smith', 'john.smith@email.com'),
  -> (102, 'Emma Wilson', 'emma.wilson@email.com'),
  -> (103, 'Michael Brown', 'michael.brown@email.com');
Query OK, 3 rows affected (0.01 sec)
Records: 3 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Members;
```

member_id	full_name	email
1	Anil Kumar	anil.kumar@example.com
2	Priya Sharma	priya.sharma@example.com
3	Ravi Verma	ravi.verma@example.com
101	John Smith	john.smith@email.com
102	Emma Wilson	emma.wilson@email.com
103	Michael Brown	michael.brown@email.com

```
6 rows in set (0.00 sec)
```

```
mysql> CREATE TABLE Loans (  
-> loan_id INT PRIMARY KEY,  
-> member_id INT,  
-> book_id INT,  
-> loan_date DATE,  
-> FOREIGN KEY (member_id) REFERENCES Members(member_id),  
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)  
-> );
```

```
Query OK, 0 rows affected (0.03 sec)
```

```
mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES  
-> (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'),  
-> (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'),  
-> (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12'),  
-> (7, 101, 3, '2025-03-01'), (8, 102, 3, '2025-03-07'),  
-> (9, 103, 1, '2025-03-15'), (10, 101, 1, '2025-04-01');
```

```
Query OK, 10 rows affected (0.01 sec)
```

```
Records: 10 Duplicates: 0 Warnings: 0
```

```
mysql> SELECT * FROM Loans;
```

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01

```
10 rows in set (0.00 sec)
```

```
mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title
-> FROM Loans l
-> INNER JOIN Members m ON l.member_id = m.member_id
-> INNER JOIN Books b ON l.book_id = b.book_id;
```

Member_Name	Book_Title
John Smith	The Alchemist
Emma Wilson	The Alchemist
Michael Brown	The Alchemist
John Smith	The Alchemist
Emma Wilson	Clean Code
John Smith	Clean Code
Michael Brown	Clean Code
Michael Brown	Atomic Habits
John Smith	Atomic Habits
Emma Wilson	Atomic Habits

10 rows in set (0.00 sec)

```
mysql> SELECT published_year, COUNT(book_id) AS Total_Books
-> FROM Books
-> GROUP BY published_year
-> ORDER BY published_year;
```

published_year	Total_Books
1988	1
2008	1
2018	1

3 rows in set (0.00 sec)

```
mysql> CREATE TABLE Donation_History (
-> donation_id INT PRIMARY KEY,
-> book_id INT,
-> donor_name VARCHAR(100),
-> donation_date DATE,
-> FOREIGN KEY (book_id) REFERENCES Books(book_id)
-> );
```

Query OK, 0 rows affected (0.03 sec)

```
mysql> START TRANSACTION;
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> INSERT INTO Books (book_id, title, isbn, published_year)
-> VALUES (4, 'Animal Farm', '9780451526342', 1945);
```

Query OK, 1 row affected (0.00 sec)

```
mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date)
      -> VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.06 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780451524935';
Empty set (0.00 sec)

mysql> |
```